Regression

Friday, September 25, 2026 8:08 AM

Classification (การจำแนกประเภท) | Regression (การถดถอย) | Clustering (การจัดกลุ่ม) | Optimization (การหาค่าที่เหมาะสมที่สุด)

Supervised Learning | Unsupervised

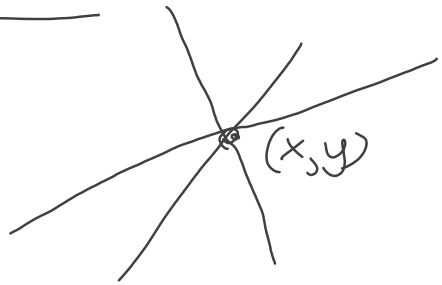
X, Y

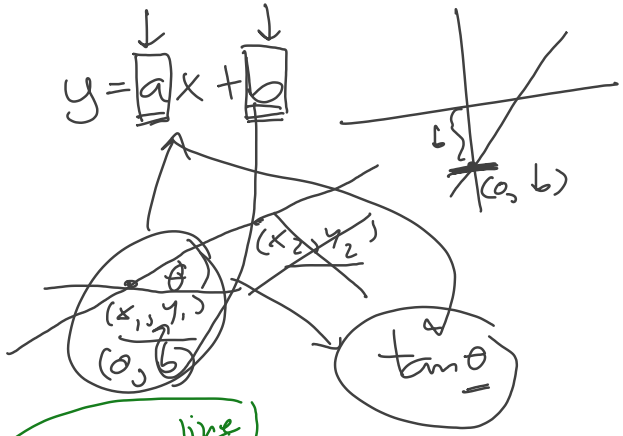
Discrete
- Y/N
- M/F
- Dog/Cat

Continuous
- Age
- Weight
- height

Linear Regression

เส้น(ตรง)





MAE / MSE

error ← metric

mean

aggregate

optimum line

Overfitting

V = IR + 0 (Ohm's Law)

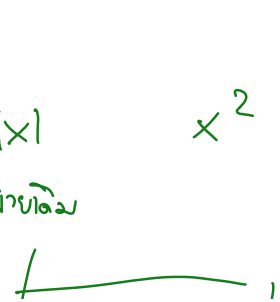
y = aX + C

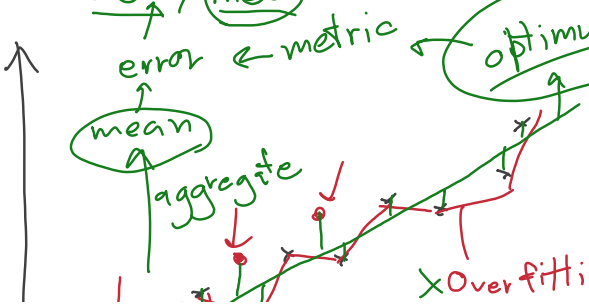
V = RI

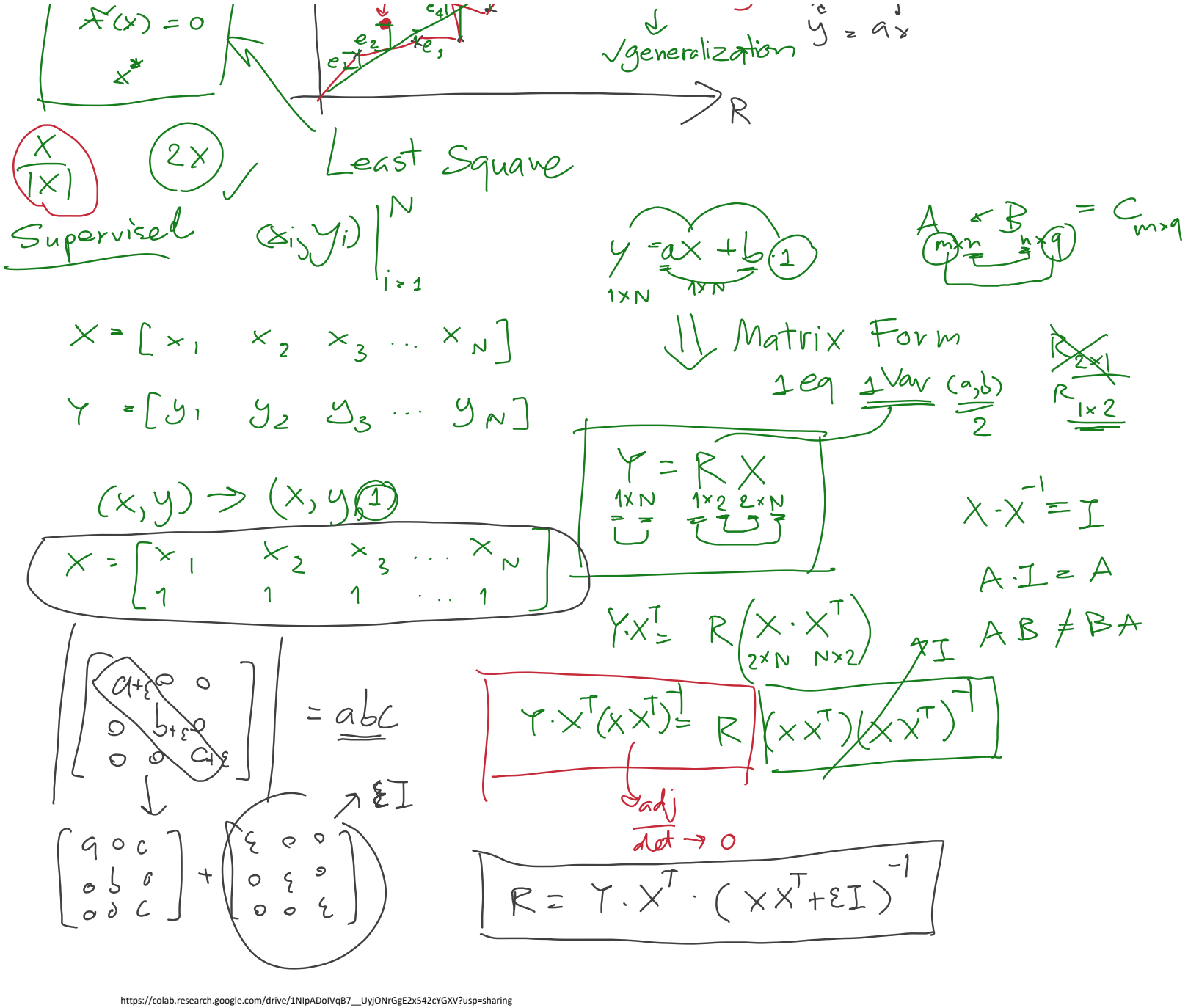
|x|

x^2

แบ่งเป็นส่วน

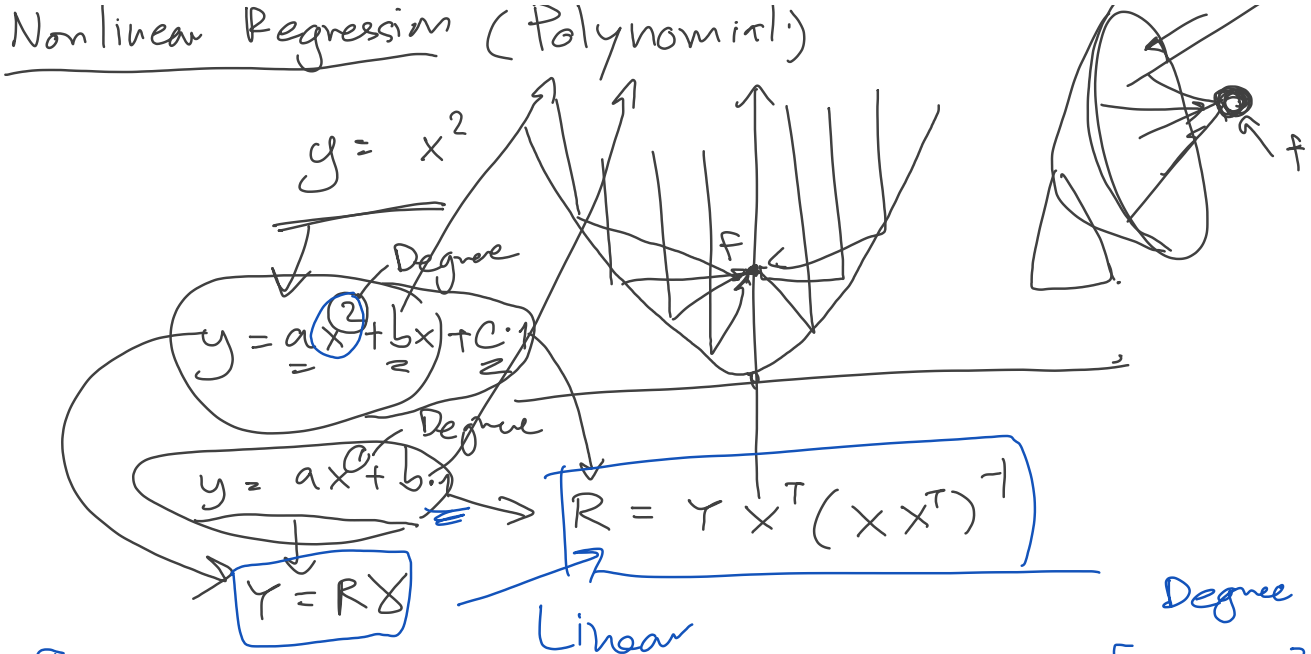






https://colab.research.google.com/drive/1NIpADoIVqB7__UyjONrGgE2x542cYGXV?usp=sharing

Nonlinear Regression (Polynomial)



Linear

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \rightarrow \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} x_1^2 & x_2^2 & x_3^2 & x_4^2 \\ x_1 & x_2 & x_3 & x_4 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Degree 2

Degree

$$y = a_1 x^n + a_2 x^{n-1} + a_3 x^{n-2} + \ldots + a_n$$

Degree 1 = Linear = เส้นตรง

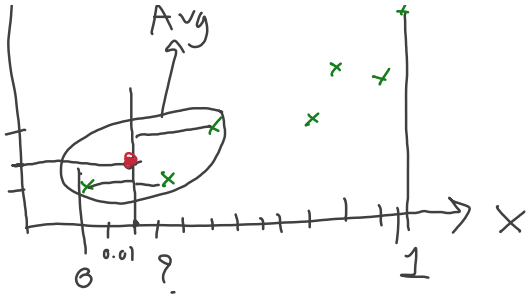
Degree 2 = Parabola = เส้น 1 ลูก

Degree 3 = เส้น 2 ลูก

Degree n = เส้น n-1 ลูก

K-NN → regressor





PM

GA

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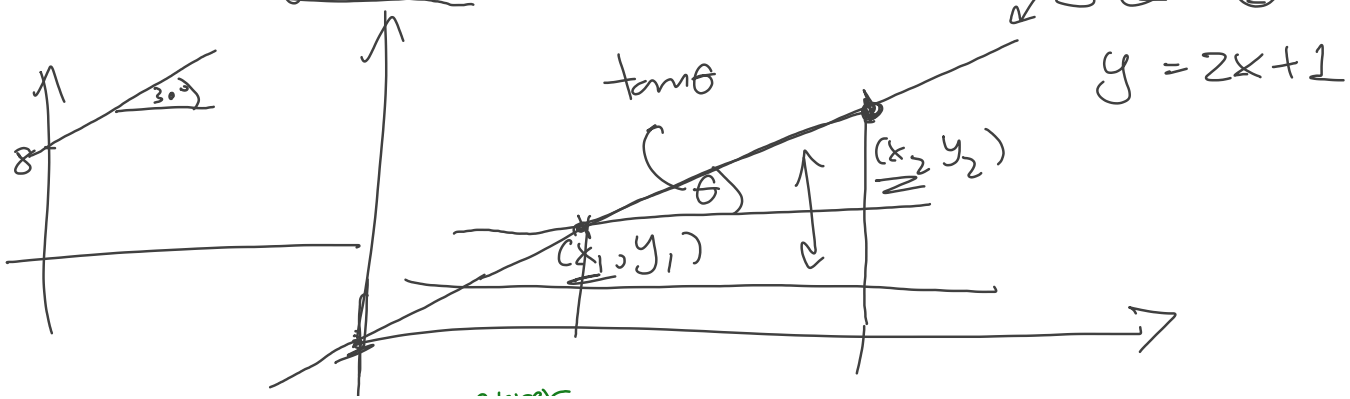
(X, Y) | (X, Y) | (X,)

Supervised Learning | Unsupervised

Discrete
- Y/N
- M/F
- Dog/Cat
- 1, 2, 3, 4

Continuous
- Age
- W/H

Linear Regression (การถดถอยเชิงเส้น)



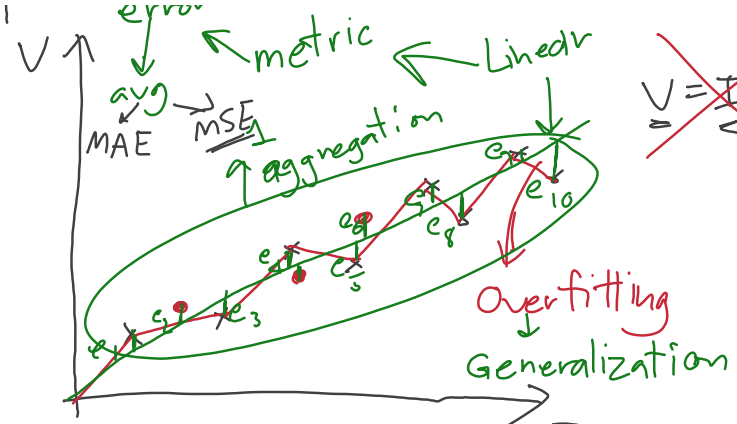
$y = ax + b$

$y = 2x + 1$

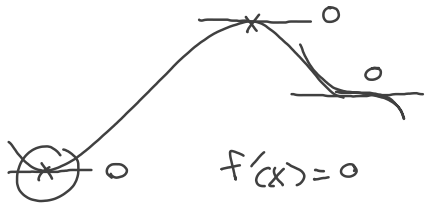
$|x|$ x^2

$f'(x) = 0$

$\frac{x}{|x|}$ $2x$



$V = IR$ (Ohm's Law)



Least Square

Matrix form

$X = [x_1 \; x_2 \; x_3 \; \cdots \; x_N]_{1\times N}$

$Y = [y_1 \; y_2 \; y_3 \; \cdots \; y_N]_{1\times N}$

$X' = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_N \\ 1 & 1 & 1 & & 1 \end{bmatrix}_{2\times N}$

$y = ax + b \cdot 1$ $\quad a, b = ?$

x' ← Homogenous Coordinates

$[y_1 \; y_2 \cdots y_N] = [a \; b] \begin{bmatrix} x_1 & x_2 & \cdots & x_N \\ 1 & 1 & & 1 \end{bmatrix}$

Y $\quad R$ $\quad X$

$1\times N$ $\quad 1\times 2$ $\quad 2\times N$

$Y = RX$

~~$Y \cdot X^{-1} = R X X^{-1}$~~

$YX^T = RXX^T$

$YX^T(XX^T)^{-1} = R$ $\quad XX^T(XX^T)^{-1}$

$2\times N$ $\quad N\times 2$

$A \cdot A^{-1} = I$

$A \cdot I = A$

$A \cdot B \neq B \cdot A$

$A^{-1} = \frac{\text{adj } A}{\det A}$

square

https://colab.research.google.com/drive/1fiU0fb4Sgd06yFp3WTb4490ygiKMcWYZ?usp=sharing

Nonlinear Regression (Polynomails)

Degree 1

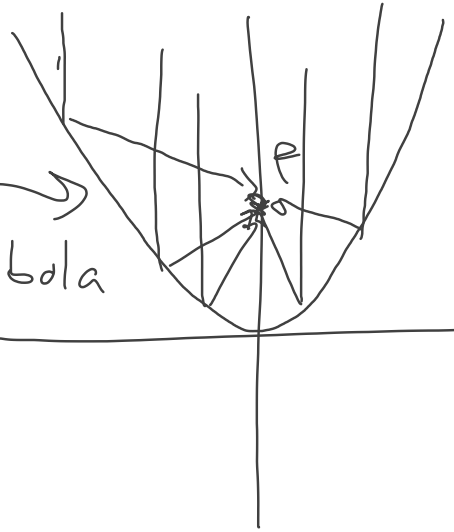
$y = ax + b$ ①

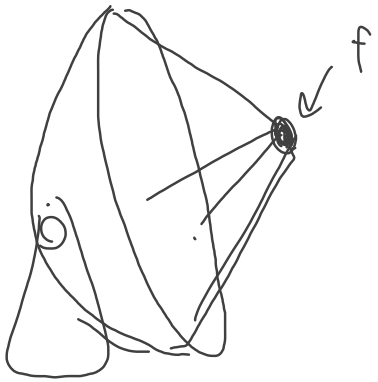
Degree 2

$y = x^2$

$y = ax^2 + bx + c$ ②

Parabola

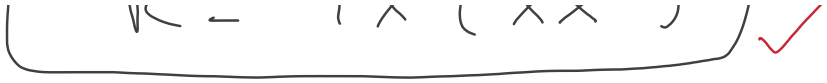




$$[y_1 \; y_2 \; y_3 \cdots y_N] = [a \quad b \quad c] \begin{bmatrix} x_1^2 & x_2^2 & \cdots & x \\ x_1 & x_2 & \cdots & \\ 1 & 1 & \cdots & \end{bmatrix}$$

Y (1×N) $= R$ (1×3) $\times X$

$$Y = RX$$



Degree 1

$$y = ax + b$$

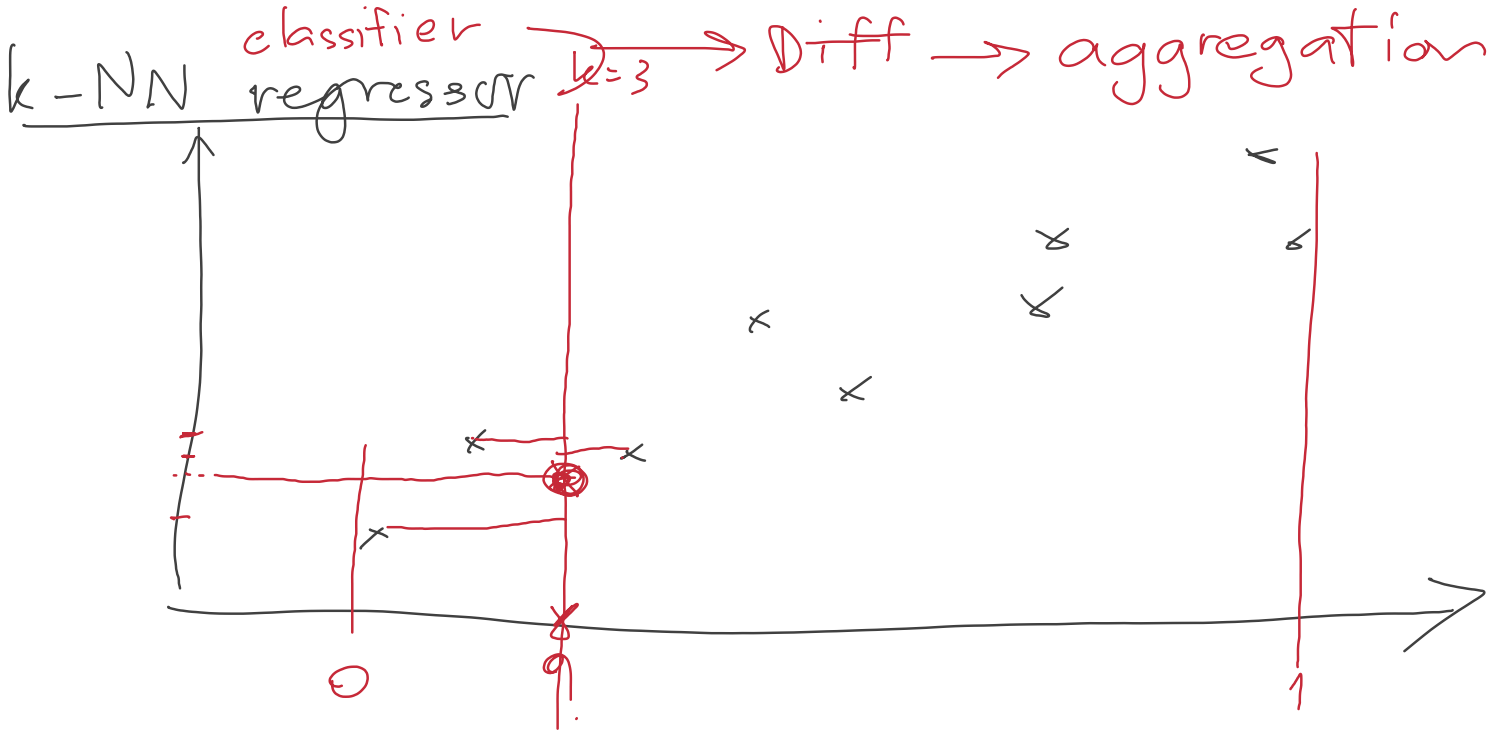
Degree 2

$$y = ax^2 + bx + c$$

$$\vdots$$

$$y = a_1 x^n + a_2 x^{n-1} + \dots + a_n$$

Degree = n $\rightarrow$ $n-1$


k-NN
classifier
regressor
k=3
Diff
aggregation
0
1